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0.6-1 | Eigenvalues and Vectors

Analyze a 3-story shear frame using the flexibility method to determine natural frequencies and mode shapes (fter Clough and Penzien [0.6.1])

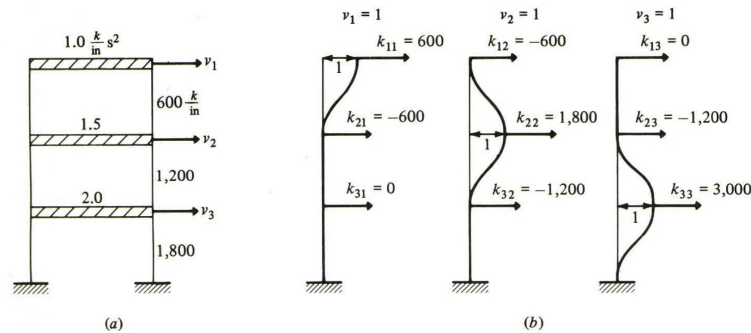


Fig. 1 - Structural Model

```
import numpy as np
import numpy.linalg as la
import textwrap as tw
# set up mass and stiffness arrays
m = np.array([[1.0,0,0],[0,1.5,0],[0,0,2.0]],float)
k1 = 600*np.array([[1,-1,0.0],[-1,3,-2],[0,-2,5]],float)
# flexibility and dynamic matrix
f = la.inv(k1)
d = np.inner(f,m)
eigen = la.eig(d)
evalus = eigen[0]
print("\neigenvalues:\n", " "*8,evalus)

eigenvalues:
      [0.00474206  0.00103739  0.00047055]
#
nat_freq = 1/(np.sqrt(evalus))
print("\nnatural frequencies:\n", " "*8,nat_freq)

natural frequencies:
      [14.52166783  31.04769646  46.09947622]
#
evect = np.array(eigen[1])
print("\neigenvectors:\n",tw.indent(str(evect), " "*8))

eigenvectors:
      [[-0.81332769 -0.73942881  0.27304451]
       [-0.52747169  0.44853685 -0.69406171]
       [-0.24550292  0.50205551  0.66612689]]
```

0.6-2 | Plot Mode Shapes

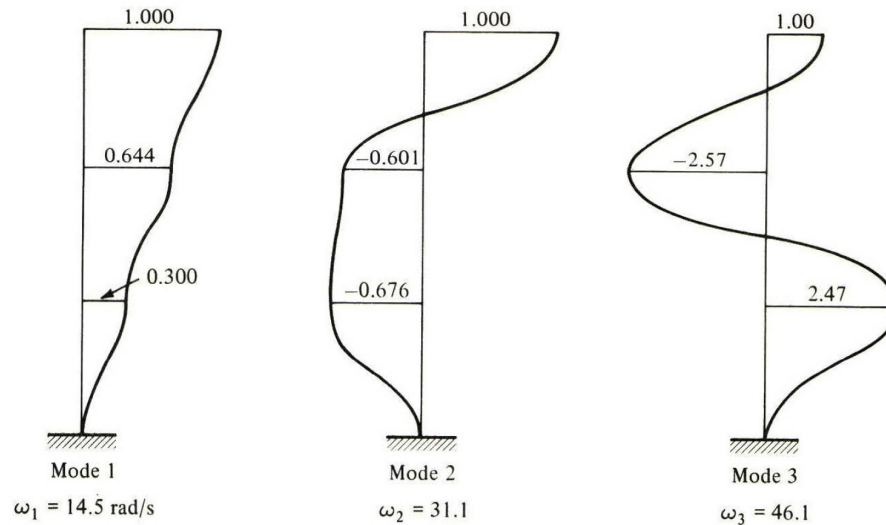


Fig. 2 - Structural Model

Plot normalized mode shapes and compare to Penzien and Clough. [0.6.2]

```
import numpy as np
import matplotlib.pyplot as plt
import os
from tabulate import tabulate as tb
# initialize eigenvector array (need (x,1) shapes for plotting)
ms = np.shape(evect)
zz = np.zeros((ms[0],1))
x1=np.concatenate((evect,zz),1)
#plot mode shapes using matplotlib
y=np.array([0,1,2,3])
m3=x1[2]*.35+5
m2=x1[1]*.35+3
m1=x1[0]*.35+1
m=np.concatenate((m1,m2,m3))
plt.clf()
plt.plot(m1,y)
plt.plot(m2,y)
plt.plot(m3,y)
plt.xlim(.5,6.)
plt.xlabel('mode')
plt.ylabel('levels')
plt.title("Mode Shapes")
plt.grid()
curdir=os.getcwd()
imgdir=os.path.join(curdir,"img","mode_shapes.png")
plt.savefig(imgdir)
# table of eigenvalues and normalized eigenvectors
```



```

evec_tt = np.transpose(evec)
for i in range(len(nat_freq)):
    evec_tt[i] = evec_tt[i]/evec_tt[i][0]
xx = np.concatenate((nat_freq[:, np.newaxis], evec), 1)
xx = np.round(xx, 4)
yy = ["freq", "level 3", "level 2", "level 1"]
tt = np.vstack((yy, xx))
print("\nTable of eigenvalues and normalized eigenvectors\n",
      tb(tt, headers="firstrow", tablefmt="rst"))

```

Table of eigenvalues and normalized eigenvectors

freq	level 3	level 2	level 1
14.5217	1	1	1
31.0477	0.6485	-0.6066	-2.5419
46.0995	0.3018	-0.679	2.4396

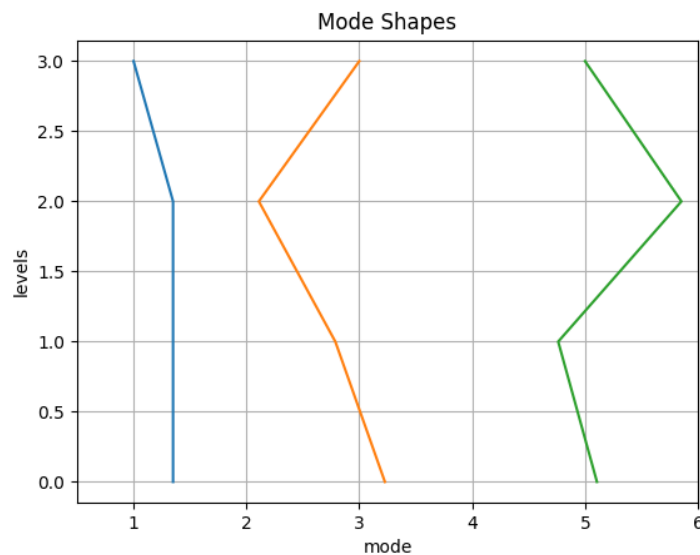


Fig. 3 - Calculated Normalized Modes

[0.6.1]

R.W. Clough and J. Penzien, Dynamics of Structures. New York, NY, USA:McGraw-Hill, 1975. pg. 178-180

[0.6.2]

ibid. pg. 180-182

